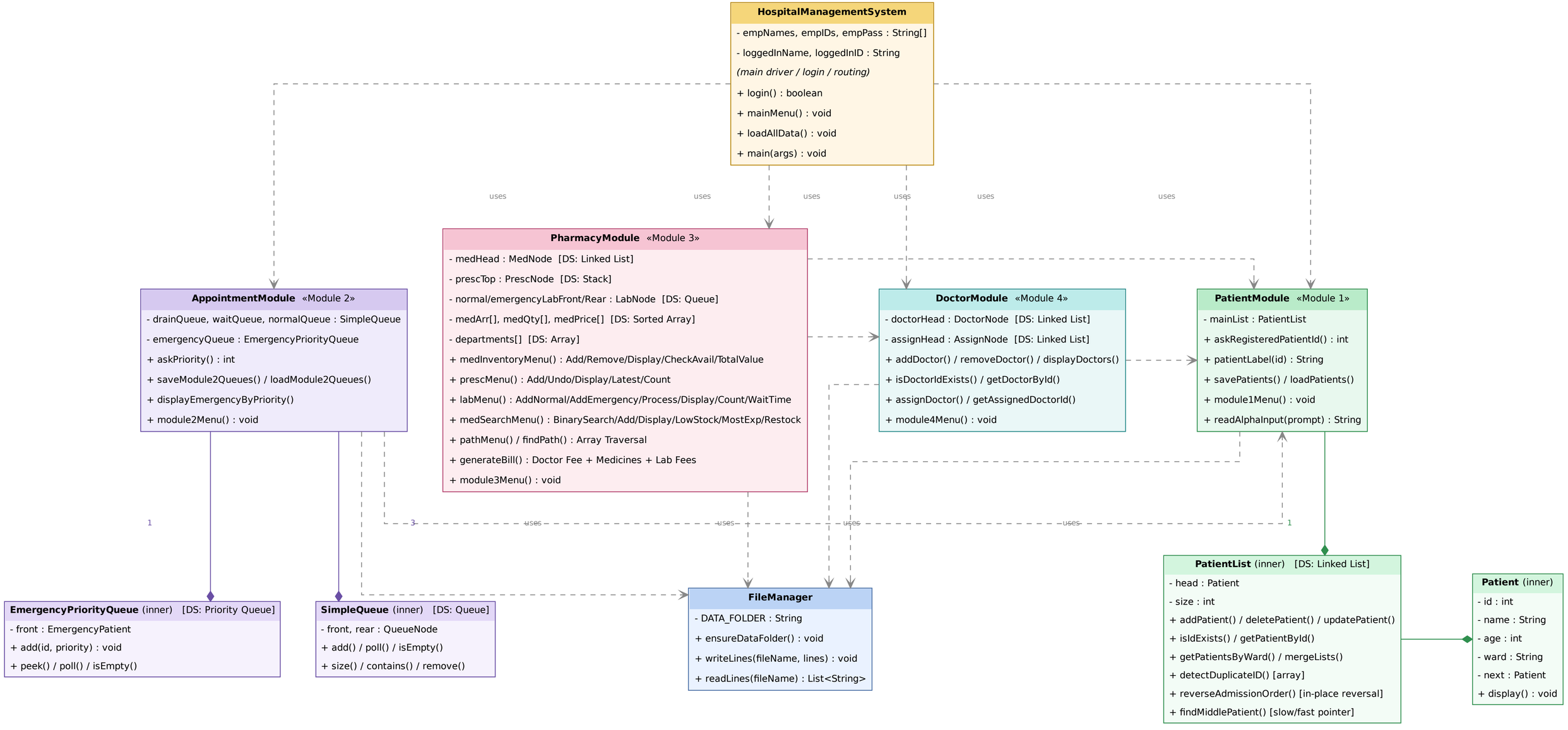
HospitalManagementSystem
- empNames, empIDs, empPass : String[]
- loggedInName, loggedInID : String
(main driver / login / routing)
+ login() : boolean
+ mainMenu() : void
+ loadAllData() : void
+ main(args) : void
uses
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uses
PharmacyModule «Module 3»
- medHead : MedNode [DS: Linked List]
- prescTop : PrescNode [DS: Stack]
- normal/emergencyLabFront/Rear : LabNode [DS: Queue]
- medArr[], medQty[], medPrice[] [DS: Sorted Array]
- departments[] [DS: Array]
+ medInventoryMenu() : Add/Remove/Display/CheckAvail/TotalValue
+ prescMenu() : Add/Undo/Display/Latest/Count
+ labMenu() : AddNormal/AddEmergency/Process/Display/Count/WaitTime
+ medSearchMenu() : BinarySearch/Add/Display/LowStock/MostExp/Restock
+ pathMenu() / findPath() : Array Traversal
+ generateBill() : Doctor Fee + Medicines + Lab Fees
+ module3Menu() : void
AppointmentModule «Module 2»
- drainQueue, waitQueue, normalQueue : SimpleQueue
- emergencyQueue : EmergencyPriorityQueue
+ askPriority() : int
+ saveModule2Queues() / loadModule2Queues()
+ displayEmergencyByPriority()
+ module2Menu() : void
DoctorModule «Module 4»
- doctorHead : DoctorNode [DS: Linked List]
- assignHead : AssignNode [DS: Linked List]
+ addDoctor() / removeDoctor() / displayDoctors()
+ isDoctorIdExists() / getDoctorById()
+ assignDoctor() / getAssignedDoctorId()
+ module4Menu() : void
PatientModule «Module 1»
- mainList : PatientList
+ askRegisteredPatientId() : int
+ patientLabel(id) : String
+ savePatients() / loadPatients()
+ module1Menu() : void
+ readAlphaInput(prompt) : String
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uses
uses
uses
uses
1
EmergencyPriorityQueue (inner) [DS: Priority Queue]
- front : EmergencyPatient
+ add(id, priority) : void
+ peek() / poll() / isEmpty()
SimpleQueue (inner) [DS: Queue]
- front, rear : QueueNode
+ add() / poll() / isEmpty()
+ size() / contains() / remove()
FileManager
- DATA_FOLDER : String
+ ensureDataFolder() : void
+ writeLines(fileName, lines) : void
+ readLines(fileName) : List<String>
PatientList (inner) [DS: Linked List]
- head : Patient
- size : int
+ addPatient() / deletePatient() / updatePatient()
+ isIdExists() / getPatientById()
+ getPatientsByWard() / mergeLists()
+ detectDuplicateID() [array]
+ reverseAdmissionOrder() [in-place reversal]
+ findMiddlePatient() [slow/fast pointer]
Patient (inner)
- id : int
- name : String
- age : int
- ward : String
- next : Patient
+ display() : void